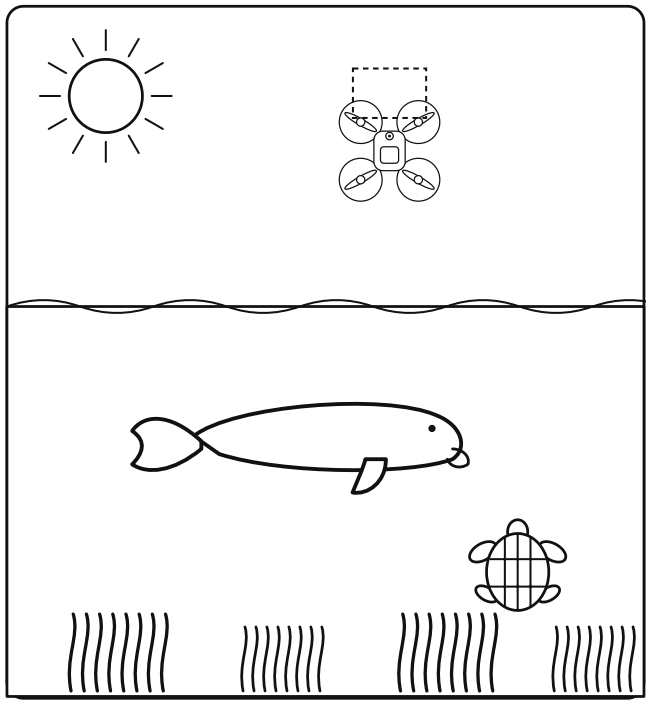
Printable pack
Level 6 · Year 9 to 10
No internet needed

AI FIRST NATIONS · DRONE KNOWLEDGE ON COUNTRY

NGARRAKAN NGARDIJI

LEVEL 6 · AI ON COUNTRY · DUGONG AND THE COUNTING MACHINE



1 · Teacher guide and pathway
2 · Story to read aloud
3 · Rules
4 · Colour & Create
5 · Match-up
6 · Quiz
7 · Quiz answers
8 · Certificate

Created by John Fejo (Pop Pop), Founder & Chief Remote Pilot, AI First Nations · CASA ReOC CASA.ReOC.8979 · Darwin, Larrakia Country. Goes with Drone Class 401.

PAGE 1 · FOR TEACHERS AND ELDERS

LEVEL 6 TEACHER GUIDE

Level 6: AI on Country. Kiki and KK survey seagrass, and an AI counts dugong feeding trails. Dugong teaches them that the AI can be wrong, so the rangers check every count. This level teaches AI, data and the CARE principles.

Time	What happens	Page
5 min	Acknowledgement of Country. Invite an Elder or ranger to share the local names for the animals in today's story.	—
10 min	Read the story aloud, or play it on a screen. Stop at each scene and yarn about it.	2
10 min	Rules. Read them together and link each rule to the story.	3
15 min	Colour & Create, and the match-up.	4, 5
10 min	Quiz, marked together.	6, 7
10 min	Certificate. If flying is planned, run the pre-flight checklist with the instructor.	8

THE PATHWAY: SAME SIX STEPS, YEAR 3 TO YEAR 12

Every level: **1 Story · 2 Rules · 3 Colour & Create · 4 Match-up · 5 Quiz · 6 Certificate.**

Lvl	Years	Level and story	CASA focus
1	Year 3	**First Flight** The Pilot and the Spotter	Keep the drone in sight · no higher than 120 m · 30 m from people · daylight only · never over people.
2	Year 4	**Safe Flight** Storm Coming	Weather and wind limits · battery reserve · Return to Home · lost signal · keep clear of emergencies.
3	Year 5	**Eyes in the Sky** Crocodile Says Keep Back	Keep distance from wildlife · privacy · no filming people without asking · camera settings with a purpose.
4	Year 6	**Mapping Country** Goondoi's Map	Planned missions · overlap · battery planning · instructor holds the controls for automated flights.
5	Year 7 to 8	**Sharing the Sky** Fire on the Hill	Airports and 5.5 km · CASA-verified drone safety apps · NOTAMs · registration · give way to crewed aircraft.
6	Year 9 to 10	**AI on Country** Dugong and the Counting Machine	Data from commercial flights under a ReOC · storing data safely · human checks on AI results.
7	Year 11 to 12	**Careers in the Sky** Kiki Earns Her Wings	Operator accreditation (16+) · RePL · ReOC · registration · job safety analysis · client briefs.

Flights in AI First Nations classes run under ReOC CASA.ReOC.8979 with the instructor in charge. Students under 16 always fly under supervision. Culture: CARE principles: data from Country is owned and controlled by Traditional Owners.

PAGE 2 · STORY

DUGONG AND THE COUNTING MACHINE

Read this aloud to the class. On a screen, the same story plays as an animation.

1. Kiki flies a survey over the seagrass for the sea rangers. She flies high and steady, so Dugong and Turtle aren't disturbed.
2. Back at base, an AI program looks at the photos and draws a box around every dugong it finds. It counts three.
3. Dugong laughs: "One of those is a rock!" The AI has made a mistake. This is called a false positive.
4. KK does the ranger check. She looks at every box and fixes the count: two dugong and one rock.
5. Turtle asks who the photos belong to. Pop Pop explains the CARE principles: data from Country is owned and controlled by its Traditional Owners.
6. The results go to the community first, so they can protect the seagrass where dugong feed. Ngarrakan Ngardiji: see far, check carefully, care for Country.



Yarn about it: How did KK catch the AI's mistake? Why should the community see the results first? What else could AI help rangers count on your Country?

AI ON COUNTRY

1 Fly with a purpose

Plan exactly what data you need before you fly. Don't collect more than you need.

2 Data from Country belongs to Country

Under the CARE principles, Traditional Owners own and control data about their Country.

3 Store data safely

Use approved, secure storage. Never leave data on a shared or public computer.

4 No public AI tools without permission

Never upload images of Country or people to public AI websites.

5 AI makes mistakes

A ranger checks every AI result before anyone uses it.

6 Label honestly

AI learns from labelled examples. Careful, honest labels make better AI.

7 Test on new images

Test the AI on photos it has never seen, to find out how good it really is.

8 Share results with community first

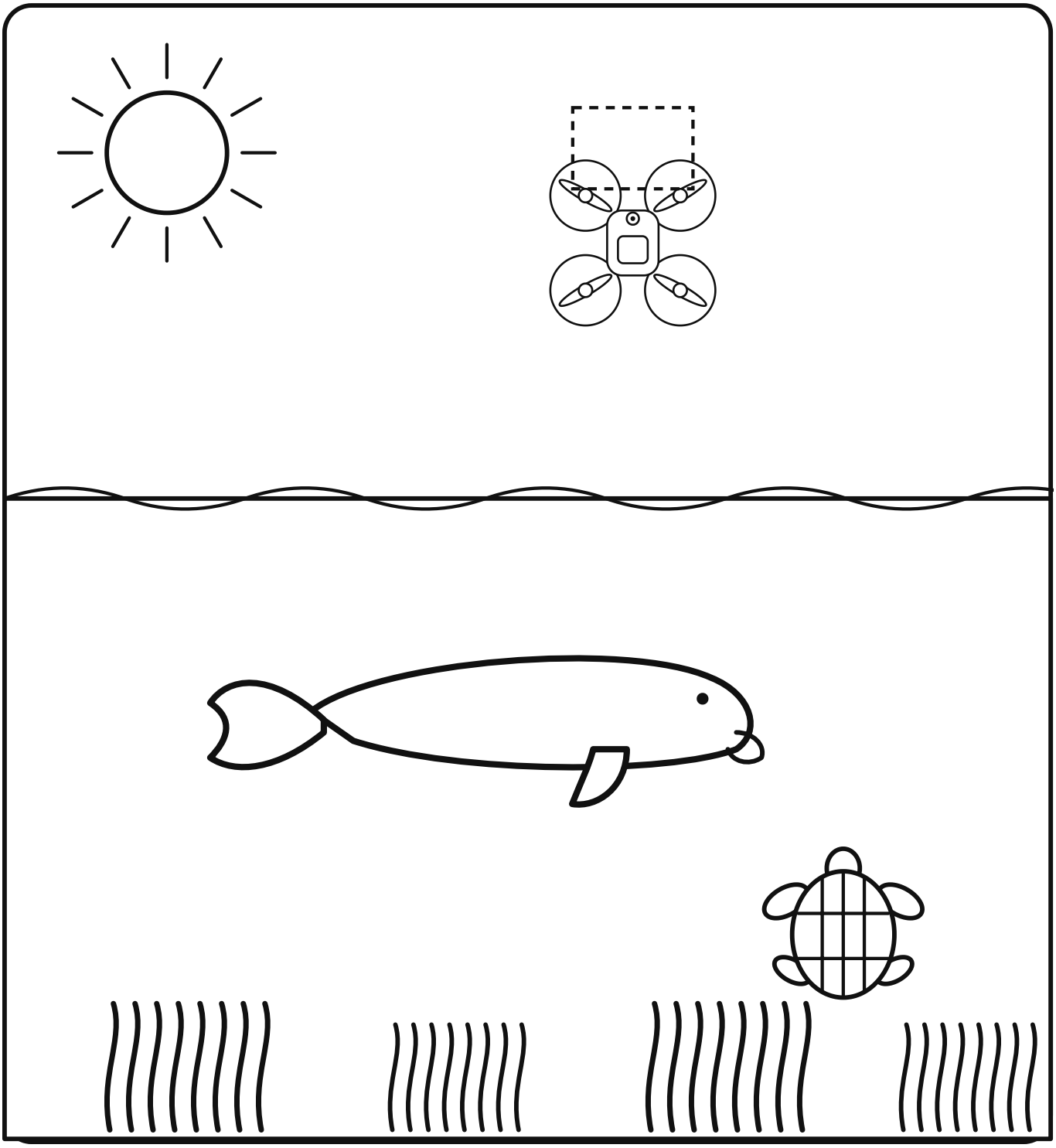
Communities see their results first and decide what happens next.

Rules can change. Check casa.gov.au and a CASA-verified drone safety app before every flight. AI First Nations flights run under ReOC CASA.ReOC.8979.

PAGE 4 · COLOUR & CREATE

DUGONG SURVEY

Name: _____ Year: _____



Colour the picture with the colours of your Country. **Create:** add one more animal or plant from your Country, and write its name in your language if you know it.

PAGE 5 · MATCH-UP

AI ON COUNTRY

Name: _____ Year: _____

Draw a line from each item on the left to its match on the right.

AI WORD	WHAT IT MEANS
Training data •	• How often the AI is right
Label •	• Telling the AI what is in a picture
Accuracy •	• Examples the AI learns from
False positive •	• Rules for Indigenous data: Collective benefit, Authority, Responsibility, Ethics
Ranger check •	• A person confirms every result
CARE principles •	• AI says something is there, but it isn't

Create: write or draw how today's rules help care for Country.

PAGE 6 · QUIZ

LEVEL 6 QUIZ

Name: _____ Year: _____

Tick one box for each question.

1. What does the A in CARE stand for?

☐ Artificial ☐ Aircraft ☐ Authority to control

2. The AI counts a rock as a dugong. What is that called?

☐ A false positive ☐ A true count ☐ A weather error

3. Who checks AI results before they are used?

☐ Another AI only ☐ A ranger or trained person ☐ Nobody

4. Can you upload drone photos of Country to a public AI website?

☐ Yes, always ☐ Only on weekends

☐ Not without permission from Traditional Owners

5. What is training data?

☐ Labelled examples the AI learns from ☐ A fitness program

☐ The drone's battery level

6. Why test the AI on new photos?

☐ No reason ☐ To see how well it really works ☐ To make it slower

7. Who should see survey results first?

☐ Social media ☐ Nobody ☐ The community and Traditional Owners

8. Where should survey data be stored?

☐ Approved, secure storage ☐ A shared library computer

☐ On a USB left in the car

9. Why plan exactly what data you need?

☐ It's not important ☐ So you only collect what is needed

☐ To fill the memory card

10. Dugong feeding trails show up in seagrass. Why does that matter?

☐ They are only pretty ☐ They show where boats go

☐ They show where dugong feed, so the seagrass can be protected

QUIZ ANSWERS

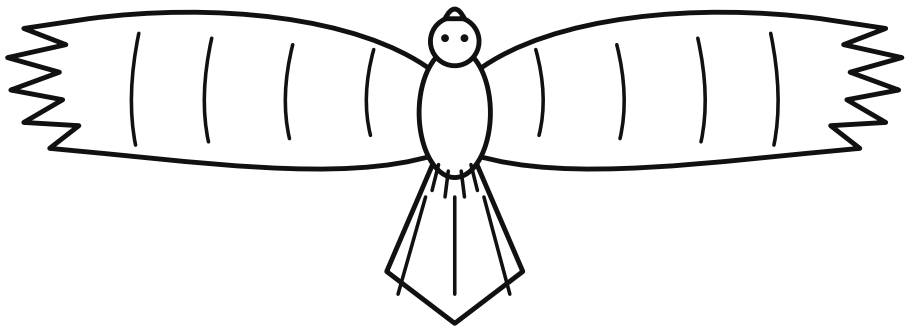
1	Authority to control
2	A false positive
3	A ranger or trained person
4	Not without permission from Traditional Owners
5	Labelled examples the AI learns from
6	To see how well it really works
7	The community and Traditional Owners
8	Approved, secure storage
9	So you only collect what is needed
10	They show where dugong feed, so the seagrass can be protected

Match-up answers: Training data → Examples the AI learns from · Label → Telling the AI what is in a picture · Accuracy → How often the AI is right · False positive → AI says something is there, but it isn't · Ranger check → A person confirms every result · CARE principles → Rules for Indigenous data: Collective benefit, Authority, Responsibility, Ethics.

AI FIRST NATIONS · DRONE KNOWLEDGE ON COUNTRY

NGARRAKAN NGARDIJI

LEVEL 6 · AI ON COUNTRY



This certifies that

_____

has completed Level 6 of Ngarrakan Ngardiji: AI, data and the CARE principles, following the Civil Aviation Safety Authority (CASA) drone safety rules.

School or community _____ Date _____

John Fejo, Founder & Chief Remote Pilot
AI First Nations · CASA ReOC
CASA.ReOC.8979

Teacher, Elder or ranger